

Rigid-Body DIC Renderer Convergence: Analysis and Conclusions

Detailed analysis and conclusions

- **The analytic-reference RMSE is the appropriate primary metric.** The quantity `RmseAnalRefUx/Uy` directly measures the pointwise change in the DIC displacement field caused by the rendering approximation. This is more revealing than `RmseExactDisp`, because the latter contains the intrinsic DIC error and can improve when renderer error accidentally cancels DIC error.
- **A renderer can look better against the prescribed displacement while actually being further from the correct DIC measurement.** The clearest example is Gaussian speckles with B-spline at $OS = SSAA = 1$. Its total DIC error against the imposed displacement is only about 1.2×10^{-3} px, smaller than the analytic result at about 1.8×10^{-3} px, yet its displacement field differs from the analytic-render DIC field by about

$$\boxed{1.57 \times 10^{-3} \text{ px RMSE}}.$$

The apparently improved DIC performance is therefore artificial cancellation or smoothing, not renderer accuracy.

- **The B-spline 1×1 “zero-bias” resonance is decisively exposed by the pointwise RMSE.** Its mean renderer-induced displacement difference is relatively modest, while the spatially varying component is large. For 12-bit Gaussian speckles, the analytic-reference RMSE is 1.57×10^{-3} px, with the difference-field standard deviation around 1.38×10^{-3} px. The near-zero conventional mean bias therefore hides a substantial spatially varying renderer error.
- **Spatial scatter matters, but the analytic-reference RMSE already captures it in a single robust measure.** For Gaussian speckles at fixed $OS = 1$, increasing SSAA makes the mean renderer-induced displacement difference almost vanish (only a few 10^{-5} px), while the pointwise RMSE plateaus near

$$\boxed{6.7 \times 10^{-4} \text{ px}}$$

for both B-spline and Catmull–Rom. Almost all of this residual is spatially varying error rather than mean bias. This explains why a traditional rigid-body bias curve can look converged while the displacement field is not.

- **Rossi-style convergence of the mean rigid bias at $OS = 1$ does not imply convergence of the DIC measurement field.** For smooth Gaussian speckles, high SSAA accurately integrates the $OS = 1$ reconstructed texture and happens to recover nearly the correct *average* displacement, but the local DIC values remain systematically different from those obtained from the analytic image. This is one of the strongest findings in the dataset.
- **Fixing $SSAA = 1$ is clearly insufficient even if OS is made arbitrarily large.** For 12-bit Gaussian speckles, B-spline plateaus near 1.52×10^{-3} px analytic-reference RMSE as $OS \rightarrow 32$, while Catmull–Rom plateaus around 1.79×10^{-3} px. Increasing texture resolution cannot compensate for an unconverged camera-pixel integral.

- **Texture representation and pixel integration are distinct numerical errors.** The two limiting processes behave differently:

$$OS \rightarrow \infty, \quad SSAA = 1$$

converges to an inadequately integrated pixel value, while

$$OS = 1, \quad SSAA \rightarrow \infty$$

accurately integrates the wrong reconstructed texture. Only simultaneous refinement of both axes can approach the analytic image-formation limit.

- **For smooth Gaussian speckles, simultaneous OS/SSAA refinement gives convincing convergence to the analytic DIC result.** At 12 bit, the diagonal analytic-reference RMSE for B-spline falls approximately as

$$1.57 \times 10^{-3}, 5.61 \times 10^{-4}, 1.86 \times 10^{-4}, 1.02 \times 10^{-4}, 9.3 \times 10^{-5}, 5.8 \times 10^{-5} \text{ px}$$

for $OS = SSAA = 1, 2, 4, 8, 16, 32$.

- **Catmull–Rom starts dramatically worse but converges rapidly once sufficiently resolved.** For the same Gaussian case, its diagonal analytic-reference RMSE is approximately

$$1.12 \times 10^{-2}, 1.26 \times 10^{-3}, 1.63 \times 10^{-4}, 2.8 \times 10^{-5}, 1.1 \times 10^{-5}, 9 \times 10^{-6} \text{ px}.$$

The large low-resolution Catmull–Rom bias is therefore a sampling/ringing effect rather than evidence of a different asymptotic solution.

- **The statement that “anything above 1×1 on the diagonal collapses to analytic” is too strong.** The conventional mean-bias curves may appear to collapse that quickly, but the pointwise analytic-reference RMSE shows residual error. B-spline still has about 5.6×10^{-4} px error at 2×2 , and even 8×8 is around 10^{-4} px. Catmull–Rom is effectively below 10^{-4} px by about 8×8 .
- **Agreement between B-spline and Catmull–Rom at high resolution is strong evidence for a common continuum limit.** The two kernels have radically different low-resolution behaviour and reconstruction properties, yet both approach the analytic DIC result as OS and SSAA are refined. This directly supports using high-OS/high-SSAA Riley calculations as numerical references for later affine and finite-star cases where an analytic rendered reference is unavailable.
- **Sharp additive disks are a substantially more severe convergence problem.** At 12 bit, B-spline diagonal analytic-reference RMSE decreases roughly as

$$1.15 \times 10^{-2} \rightarrow 5.35 \times 10^{-3} \rightarrow 2.20 \times 10^{-3} \rightarrow 7.7 \times 10^{-4} \rightarrow 2.5 \times 10^{-4} \rightarrow 9.6 \times 10^{-5} \text{ px},$$

from 1×1 to 32×32 . The discontinuity therefore requires much greater resolution than the smooth Gaussian field.

- **Catmull–Rom is much worse than B-spline for severely under-resolved disks, as expected from ringing, but ultimately converges faster.** Its disk analytic-reference RMSE begins at a very large 4.28×10^{-2} px at 1×1 , then falls to approximately

$$6.0 \times 10^{-3}, 1.14 \times 10^{-3}, 2.42 \times 10^{-4}, 7.7 \times 10^{-5}, 5.2 \times 10^{-5} \text{ px}$$

along the diagonal. B-spline is therefore safer at very coarse resolution because smoothing suppresses ringing, whereas Catmull–Rom gives a rapidly convergent approximation once the edge is adequately sampled.

- **For sharp disks, high SSAA with $OS = 1$ can make the DIC result progressively worse.** For B-spline, the analytic-reference RMSE goes from 1.15×10^{-2} at 1×1 , briefly improves at 1×2 to 4.55×10^{-3} , then increases to approximately

$$2.27 \times 10^{-2}, 3.81 \times 10^{-2}, 4.79 \times 10^{-2}, 5.35 \times 10^{-2} \text{ px}$$

as SSAA increases to 4, 8, 16 and 32. Catmull–Rom shows the same qualitative phenomenon. This is a particularly clear demonstration that converging the quadrature of an inadequately represented texture does not converge the physical image.

- **The hard-disk error at $OS = 1$ and high SSAA is predominantly spatially varying rather than a simple rigid bias.** At $OS = 1$, $SSAA = 32$, B-spline has a difference-field standard deviation around 5.2×10^{-2} px, while the magnitude of the mean difference is only around 1.5×10^{-2} px. A mean-bias plot alone therefore dramatically understates the rendering error.
- **The bespoke/function-shader SSAA sequence independently demonstrates pixel-integration convergence.** For 12-bit Gaussian speckles, analytic-reference RMSE drops from about 6.7×10^{-4} at $SSAA = 1$ to 3.1×10^{-5} at $SSAA = 32$. For sharp disks, the same progression is much slower:

$$5.98 \times 10^{-2} \rightarrow 2.36 \times 10^{-2} \rightarrow 7.98 \times 10^{-3} \rightarrow 2.73 \times 10^{-3} \rightarrow 9.74 \times 10^{-4} \rightarrow 3.49 \times 10^{-4} \text{ px.}$$

This provides a clean smooth-versus-discontinuous comparison.

- **The 8-bit results show a visible quantisation floor, particularly for B-spline.** At 32×32 , Gaussian analytic-reference RMSE is about 2.36×10^{-4} px for 8-bit B-spline versus 5.8×10^{-5} px at 12 bit. Catmull–Rom reaches about 1.3×10^{-5} versus 9×10^{-6} px. A similar effect is visible for disks. Once renderer discretisation becomes sufficiently small, image quantisation becomes the next limiting perturbation.
- **Twelve-bit data are therefore preferable for demonstrating numerical convergence.** Eight-bit results remain practically important because 8-bit DIC images are common, but quantisation can mask the asymptotic renderer behaviour. The 12-bit results expose the OS/SSAA convergence more cleanly.
- **The PSF-filtered disk cases should be treated separately unless an analytic or independently converged PSF reference is available.** Their exact-displacement statistics nevertheless show that smoothing the discontinuous disks dramatically improves DIC behaviour: the high-resolution texture cases settle around 1.7×10^{-3} px total DIC RMSE, comparable to Gaussian speckles and far below the roughly 1.16×10^{-2} px of unblurred disks.

Paper-level conclusions

1. **Synthetic-image convergence cannot be assessed from DIC bias against the prescribed displacement alone.** Renderer error can cancel intrinsic DIC error and make an unconverged image appear superior.
2. **Texture representation and pixel integration are independent convergence requirements.** Refining only one does not generally recover the analytic DIC measurement, even when conventional mean-bias curves suggest that it does.
3. **Pointwise DIC RMSE against the analytic-render reference exposes errors that are almost completely hidden by mean rigid-body bias.** In particular, the $OS = 1$, high-SSAA

Gaussian cases reproduce the mean displacement extremely well while retaining approximately 7×10^{-4} px spatially varying renderer-induced measurement error.

4. **When both OS and SSAA are refined, different reconstruction kernels converge toward the same analytic DIC solution.** This provides the key validation needed to use a demonstrably self-converged high-OS/high-SSAA Riley render as the reference for affine, finite-star and other deformation fields for which an analytic rendered reference is unavailable.